

Project Title

AI-Powered Cloud Deployment & Recovery Platform

Domain :- Cloud Computing & DevOps

Submitted By:- Himanshu Sonar

Institution:- SpeedUp Infotech

Date:- September 2026

1. Abstract

Cloud application deployment is a critical process in modern IT infrastructure. However, deployment failures due to configuration issues, permission errors, service failures, and code integration problems often result in downtime and increased operational effort.

The proposed **AI-Powered Cloud Deployment & Recovery Platform** automates application deployment on AWS cloud infrastructure while providing intelligent troubleshooting and recovery capabilities. The system deploys applications from GitHub repositories to AWS EC2 instances, stores deployment data on Amazon EBS, and creates backups using Amazon S3.

A hybrid troubleshooting engine combines rule-based analysis with Gemini AI to identify deployment failures, determine root causes, and suggest corrective actions. The platform also provides automated backup and recovery mechanisms to minimize downtime and improve reliability.

This project demonstrates the integration of Cloud Computing, DevOps Automation, Linux Administration, Shell Scripting, Python Programming, and Artificial Intelligence in a single cloud-native solution.

2. Problem Statement

In modern cloud environments, application deployment often requires manual intervention for configuration, monitoring, troubleshooting, and recovery. When deployment failures occur, engineers spend significant time analyzing logs, identifying root causes, and restoring services.

Common challenges include:

- Deployment failures caused by configuration errors.
- Permission and access-related issues.
- Apache service failures.
- Git repository integration problems.
- Manual troubleshooting processes.
- Lack of automated recovery mechanisms.
- Increased downtime and reduced productivity.

There is a need for an intelligent cloud-based platform capable of automating deployment, detecting failures, providing troubleshooting recommendations, and enabling rapid recovery.

3.Objectives

Primary Objectives

- Automate application deployment on AWS.
- Reduce manual deployment effort.
- Detect and analyze deployment failures.
- Minimize service downtime.
- Implement backup and recovery mechanisms.
- Ensure secure access management using IAM.

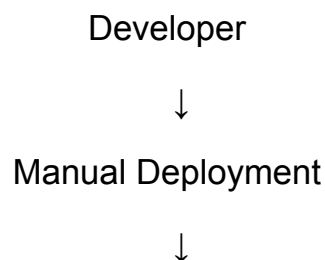
Secondary Objectives

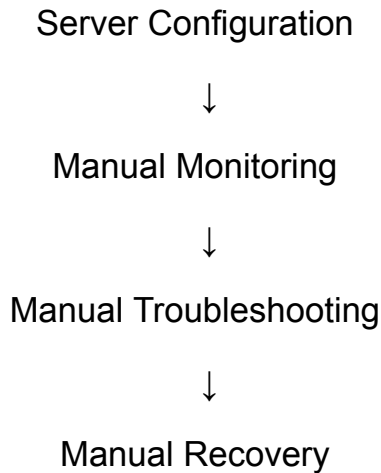
- Integrate Artificial Intelligence into cloud operations.
- Demonstrate DevOps best practices.
- Improve troubleshooting efficiency.
- Create an industry-relevant cloud project.

4.Existing System

Currently, many deployment processes are manually managed by system administrators and DevOps engineers.

Existing Workflow





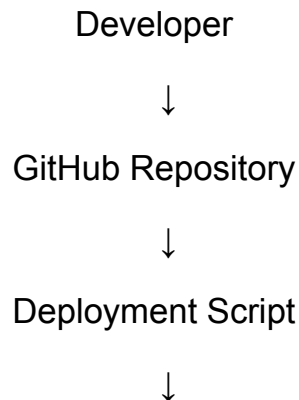
Limitations

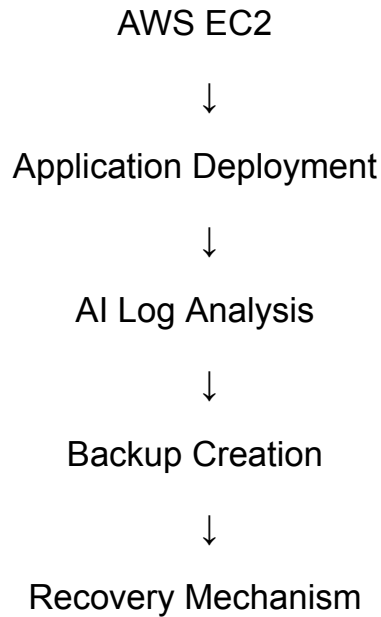
- Time-consuming deployment process.
- High dependency on technical expertise.
- Delayed fault detection.
- Manual log analysis.
- Increased operational costs.
- Higher risk of downtime.

5.Proposed System

The proposed system automates deployment and introduces intelligent troubleshooting capabilities.

Proposed Workflow





Key Features

- Automated deployment from GitHub.
- AI-powered log analysis.
- Rule-based error detection.
- Automated backup management.
- Website recovery mechanism.
- Role-based access control.

6.Innovation & Uniqueness

The proposed platform introduces several innovative features.

Hybrid AI Troubleshooting

The system combines:

Rule-Based Analysis

Detects common deployment failures:

- Permission errors
- Apache failures

- Git clone failures
- Port conflicts

Gemini AI Analysis

Provides:

- Root cause identification
- Error explanation
- Suggested corrective actions
- Troubleshooting reports

Automated Recovery

The system automatically maintains backups and enables rapid service restoration.

Complete Deployment Lifecycle

Unlike traditional deployment projects, the platform supports:

Deployment



Monitoring



Analysis



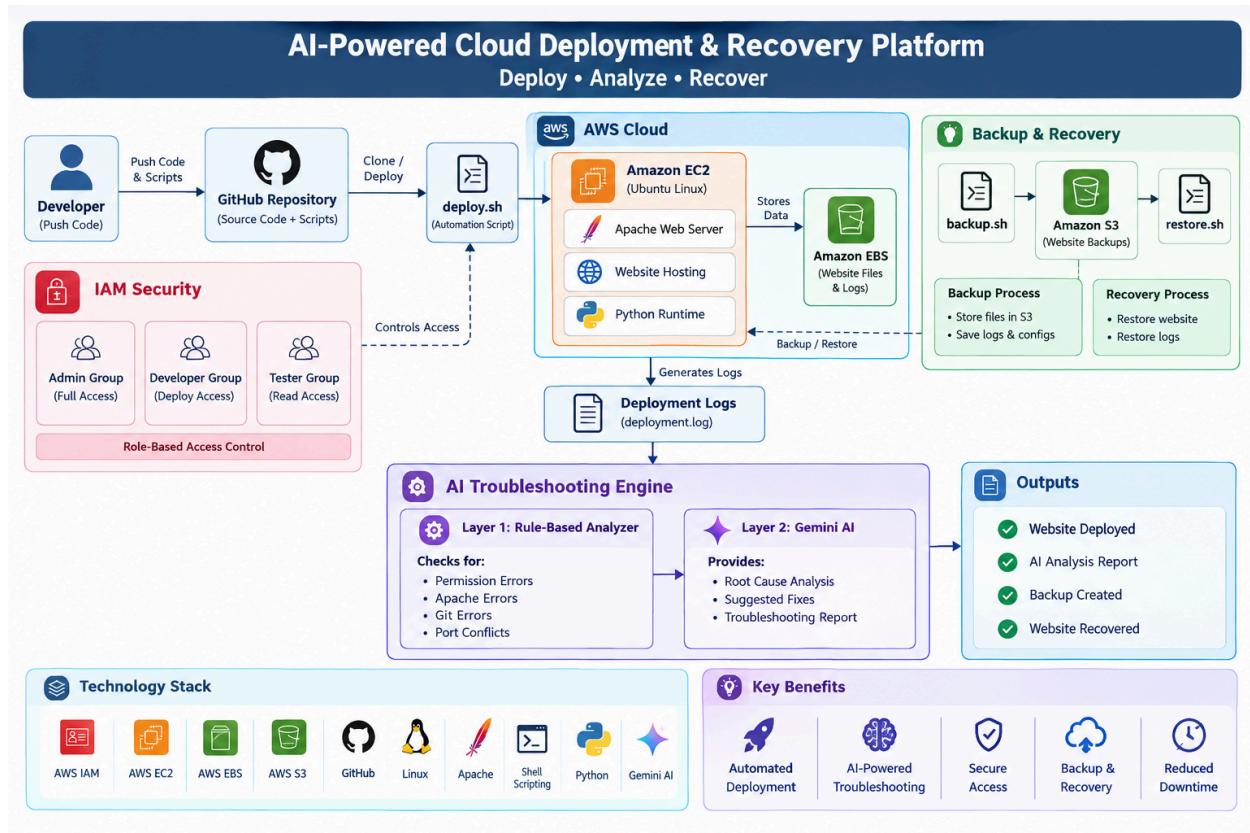
Backup



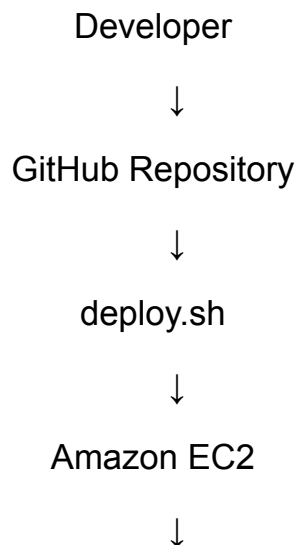
Recovery

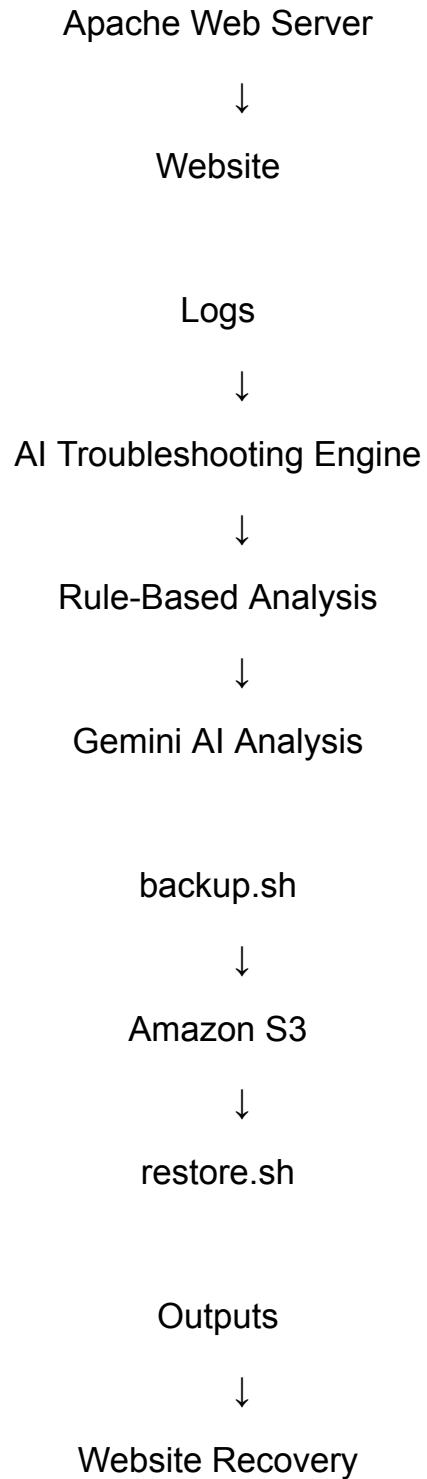
7. Architecture Diagram

System Architecture



Architecture Flow





AI Analysis Report

The architecture integrates deployment automation, AI-assisted troubleshooting, backup management, and recovery mechanisms within a secure AWS cloud environment.

8.AWS Services Used

AWS IAM

Provides Role-Based Access Control (RBAC).

Groups

- Admin Group
- Developer Group
- Tester Group

Amazon EC2

Hosts:

- Web Application
- Apache Server
- Deployment Scripts
- Python Runtime

Amazon EBS

Provides persistent storage.

Stores:

- Website Files
- Deployment Logs
- Recovery Scripts

Amazon S3

Stores:

- Website Backups
- Log Archives
- Recovery Files

9.Functional Modules

Module 1: Deployment Automation

Function

Deploys applications from GitHub to EC2.

Components

- GitHub Repository
- deploy.sh
- Apache Server

Module 2: Log Monitoring

Function

Collects deployment logs.

Logs Generated

- deployment.log
- error.log
- system.log

Module 3: AI Troubleshooting Engine

Function

Analyzes deployment failures.

Features

- Error Detection
- Root Cause Analysis
- Suggested Fixes

Module 4: Backup Management

Function

Creates automated backups.

Technology

Amazon S3

Module 5: Recovery Engine

Function

Restores website files and deployment data from backups.

10.Expected Outcome

The proposed system is expected to provide:

Technical Outcomes

- Successful application deployment.
- Faster issue identification.
- AI-generated troubleshooting reports.
- Secure cloud infrastructure.

Business Outcomes

- Reduced downtime.
- Improved deployment reliability.
- Faster recovery process.
- Better operational efficiency.

Deliverables

- ✓ Automated Deployment
- ✓ AI Analysis Report

- ✓ Backup Creation
- ✓ Recovery Mechanism
- ✓ Secure Cloud Environment

11. Future Scope

The platform can be enhanced with advanced cloud and DevOps technologies.

Docker Integration

Containerized application deployment.

Kubernetes Integration

Automated container orchestration.

CI/CD Pipeline

Integration with:

- Jenkins
- GitHub Actions
- AWS CodePipeline

Predictive Analytics

AI-based deployment failure prediction.

Multi-Cloud Deployment

Support for:

- AWS
- Microsoft Azure
- Google Cloud Platform

Self-Healing Infrastructure

Automatic corrective actions without human intervention.

12. Conclusion

The **AI-Powered Cloud Deployment & Recovery Platform** provides a modern approach to cloud deployment by combining automation, security, artificial intelligence, and disaster recovery capabilities.

The system automates application deployment, analyzes deployment failures using AI, creates backups, and enables rapid recovery of services. By integrating AWS cloud services, Linux administration, DevOps practices, and AI technologies, the platform demonstrates a practical and industry-relevant solution for reducing downtime and improving operational efficiency.

The project showcases the skills required for modern Cloud and DevOps Engineers and highlights the potential of AI-assisted cloud operations in real-world environments.